



Wibral-line PID for physical intelligence

Results, limits and the next sensor experiment

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The immediate application is **offline sensor analysis for embodied prediction**: two cameras, with optional audio, used to predict a separately measured future physical outcome. This is an attainable first step toward robust world models and robot policies. It is not yet a demonstrated training or control improvement.

The selected functional is **Makkeh–Gutknecht–Wibral categorical shared exclusions (MGW)**, in nats, with signed atoms. The defining paper is distinct from continuous shared exclusions and from other PID definitions. Fitted bins define a categorical estimand; raw sensor information is a different object.

The practical finding

An extra sensor's value is not its synergy alone. For two-source atoms R, U_1, U_2, S , the added Shannon information is $U_2 + S$. If source 1 is available with probability a and source 2 with probability b , with availability indicators M_1, M_2 such that $M_1 \perp M_2$ and $(M_1, M_2) \perp (X_1, X_2, Y)$, the added Bayes log-loss value is

$$\Delta = b[(1 - a)R + U_2 + aS].$$

Both predictors must see the same mask, and the complete-data policy and target law must remain fixed. Redundancy can provide backup value; unique information and synergy combine when sensors are present. The full proof includes correlated failures and data-dependent missingness in the detailed report.

For independent fair bits A, B, U , set the same target $Y = (A, B)$ and baseline $X_1 = A$. Candidate noise U and useful complement B both have MGW synergy $\ln(4/3)$. Their added information is respectively zero and $\ln 2$. A duplicate A has zero synergy but supplies backup value when the first copy is unavailable.



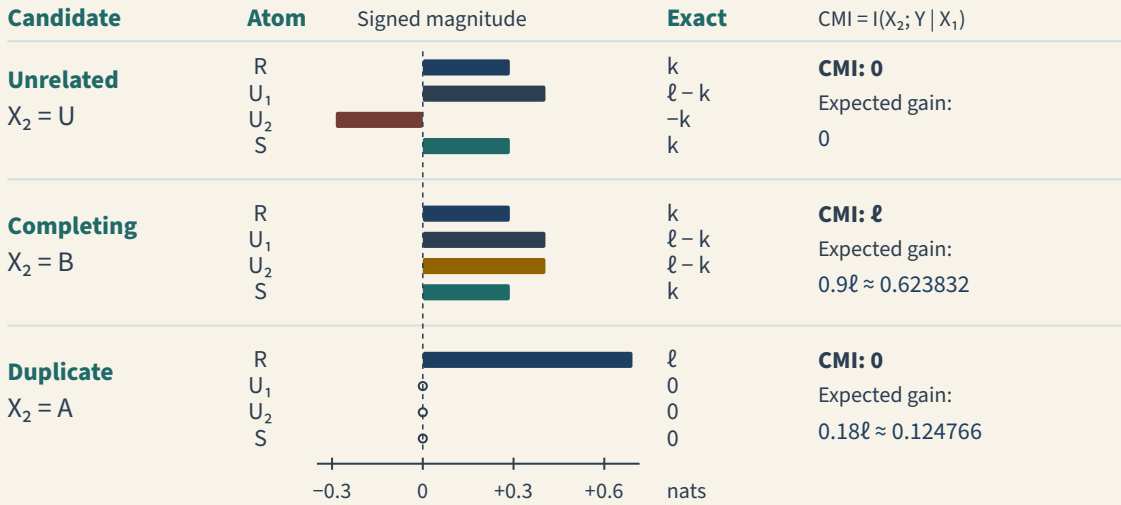


Same synergy, different added information

Exact analytic example · signed net MGW atoms in nats

A, B, U independent fair bits; target $Y = (A, B)$; baseline $X_1 = A$.

$\ell = \ln 2 \approx 0.693147$; $k = \ln(4/3) \approx 0.287682$



U and B have the same synergy k , but their added information is 0 versus $\ln 2$.

Gain: $a = 0.8$, $b = 0.9$; observed availability masks independent of data and each other.

Bayes log-loss information gain; learned-policy benefit is unproved. Zero atoms use dots.

Figure 1: One target law, three candidate sensors. Signed unique information distinguishes noise from a useful complement despite equal synergy; duplication can have backup value.

There is also a useful limit: the expected Bayes log-loss reduction relative to the no-source predictor, under a fixed law and exogenous availability, depends only on subset mutual information. For two to four sources, that is 3, 7 or 15 values. Full PID's 4, 18 or 166 atoms provide a finer allocation, but no extra decision information for that exact objective. PID must earn additional cost through a useful diagnostic question or a separately justified objective.

Why three-source MGW still matters

For two cameras and audio, the 18 signed MGW atoms describe the named measure's allocation across three declared input groups and their combinations. This can motivate a specific representation or model ablation; it does not show how a trained model uses the sensors. Seven subset MI values already determine the availability objective above, so full PID3 is an optional explanatory analysis, not a fusion or training requirement. The existing Rust calculation and its unfinished certification program are separate: stronger assurance of the allocation is useful, while calibration and practical benefit still require their own evidence. The detailed report gives two distinct camera/audio event constructions and the current closure boundary. Continuous PID3 is a different functional.





What exists and what it supports

Work	Useful role	Remaining boundary
Rust categorical MGW for 2–4 sources	Offline signed atom calculation with budgets and cancellation	No completed sensor benchmark follows from library availability
Finite-law log-score package	Four foundational and 13 conditional targets checked locally: posterior, Bayes score and MI/CMI gains	No mask/MGW formal bridge, portable or hosted replay, calibration or sensor benefit
Equal-synergy and target-copy formal work	Reject an invalid synergy-only selection rule	Exact theorem premises and replay status remain in the theorem maps
Fixed-alphabet continuity and dependence-aware law bounds	Conditional tools for estimator sensitivity	A valid sampling/dependence model is still required
Finite-prefix mean, bias and gradient work	Mathematical tools for a future bounded encoder experiment	Not an implemented robot-training method
Availability exposition	Complete classical derivation, controls and experiment design	Availability/mask and MGW allocation arguments remain handwritten; no scientific-priority claim

The [mathematical results guide](#) points to standalone proofs and exact formal status. A verified mathematical intermediate step does not establish estimator calibration or physical value.

Rust implementation and computational cost

Rust supplies budgeted categorical MGW for two to four sources through the [averaged API](#). Event calculations are quadratic in the occupied complete states K for each lattice node. Averaged output saves retained pointwise results; row and PMF storage remain. Dense log-score validation and evaluation over K_Y target labels take $O(K_Y)$ work and $O(1)$ extra storage beyond the inputs.

No Rust log-score evaluator or learner is added; floating-point execution is unverified. Feature extraction, masked prediction, training and resampling add costs. Begin offline: latency and learning benefit are unmeasured. See the [full cost discussion](#).

The ecosystem connection

CREBAIN produces observations; NCP transports sensor records; Prisoma inspects experiments; pid-rs computes information; Galadriel can retain diagnostic evidence. Manwe supplies a separate tracking setting. The detailed report distinguishes existing consumers from proposals.

The current two-camera-plus-audio example records five images and 800 pressure samples over six 120-Hz body ticks. That proves a capture composition, not independent statistical replication. The missing step is a frozen feature table joined to future reference labels. Use one declared episode/landmark unit, preserve timing and missingness, fit transforms only on training episodes, and evaluate untouched episodes.





Where the offline report is useful

The proposed camera/audio adapter processes each RGB or grayscale camera separately with a declared frozen extractor and each audio window with its own declared feature map. Low-dimensional outputs are converted to categories; each sensor is one source group. Hold shared context fixed or use an explicit conditional analysis for sensor-specific attribution. PID then describes those representations against a separately measured outcome. It does not operate directly on raw pixels or recover discarded image/audio information. A fusion predictor is trained and tested separately on the sensor subsets. The detailed report gives an explicit three-source row, distinguishes neural-representation analysis from model-use tests, and explains why a fused output cannot be relabeled as an independent sensor.

Decision	Role of the report	What still establishes success
Preparation and labeling	Specify time windows, source identity, reference outcomes, exclusions and fitted transforms.	Direct record checks and reference adjudication. PID neither cleans data nor certifies labels.
Feature and fusion design	Explain signed information allocation and nominate a matched representation or model ablation.	Held-out task loss, calibration, failure behavior and cost.
Tuning and training	Test a declared diagnostic-guided change inside development splits.	Untouched evaluation against task-only and masked-training baselines. No PID learner is supplied.
Placement and policy	Shortlist a collection or control experiment with a fixed target and explicit costs.	Candidate-location data or a justified sensor model, then task evaluation; closed-loop evidence for policy claims.
Integrity review	Compare predeclared relationships across retained regimes.	Labeled faults, benign-shift controls and calibrated alerts. Atom changes do not identify tampering.

The recorded office example makes the boundary concrete. Four training-fitted bins for light and CO₂ give positive added empirical information of about 0.001558 nats on the later recording. Yet the fixed pair predictor has worse mean log loss: 0.050762 versus 0.047443 nats per row for light alone. Its larger predictive excess loss outweighs the information gain. The detailed report gives the data statistics, predictor rule, signed cancellation and complete loss calculation.

A separately measured target is not a statistically independent target: it must retain the relationship being predicted. Split episodes before fitting or tuning, preserve natural missingness, and keep the target law fixed for the mathematical comparison. Changing physical placement or policy changes that law and requires new evidence. Compute PID offline only when its allocation answers a useful question beyond subset MI and direct task performance.

Partial visibility and distortion are separate from missing inputs. Retain quality conditions such as occlusion, blur, clipping and frame age, then compare information and prediction on common episodes within supported regimes. An offline quality label is not automatically available to the deployed predictor. Natural occlusion can depend on the target, so it does not inherit a random-dropout formula. The full report gives the conditional-law interpretation, data fields and evaluation limits.





How practical value will be tested

Start offline with all eight sensor subsets. Measure predictive loss, availability, runtime and memory. Keep fixed-model outages separate from retrained subset models. Compare MI/CMI, subset task loss and established masking methods. [Masked sensor training](#) and [MAD multi-view reinforcement learning](#) are relevant baselines; no superiority to them is established here.

The detailed report derives a conservative paired-episode loss bound and states its independence and bounded-probability assumptions. It also explains why natural occlusion needs conditional laws, why more sensors can defeat finite-data estimation, and why an action-conditioned latent must not be evaluated against its own injected action.

Testing control, placement or re-identification requires explicit target, policy, association and cost contracts. Require useful task gains against matched baselines and retain negative results.

How to cite: Sepehr Mahmoudian, *Wibral-line PID for physical intelligence*, pid-rs technical report, 2026. Include the exact repository commit or release. Use the [citation metadata](#) for the software and cite the defining method papers separately.

